



ELECTRONIC HOT AND COLD COMPRESSION DEVICE

CryothermX

Presented by 230218G 230016K 230581K 230481E

Problem Statement

Conventional thermal therapy systems, such as ice packs and hot water bags, rely on **manual preparation** and lack precise temperature control mechanisms.

As a result, people often skip hot and cold compression even when needed. These methods are prone to **rapid temperature variation**, require **repeated maintenance** such as reheating or refreezing, and present a risk of **thermal injury** due to uncontrolled exposure.

In **non-clinical environments**, such as homes, this issue is more pronounced compared to clinical settings like hospitals.

About our solution



Regulator to switch between modes in hot mode (high-red, medium-orange, low-yellow)

Adjustable hand belt for better gripping



Contact Surface

User-interactive Display

Features

- Cold compression with a default mode 8 °C-10 °C
- Hot compression with mode adjustable
 - Low Heat Mode 35 °C
 - Medium Heat Mode 41 °C
 - High Heat Mode 48 °C

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How it works?

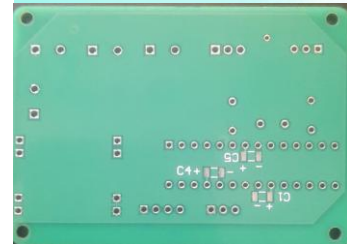
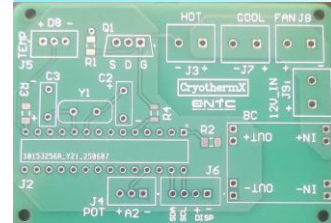
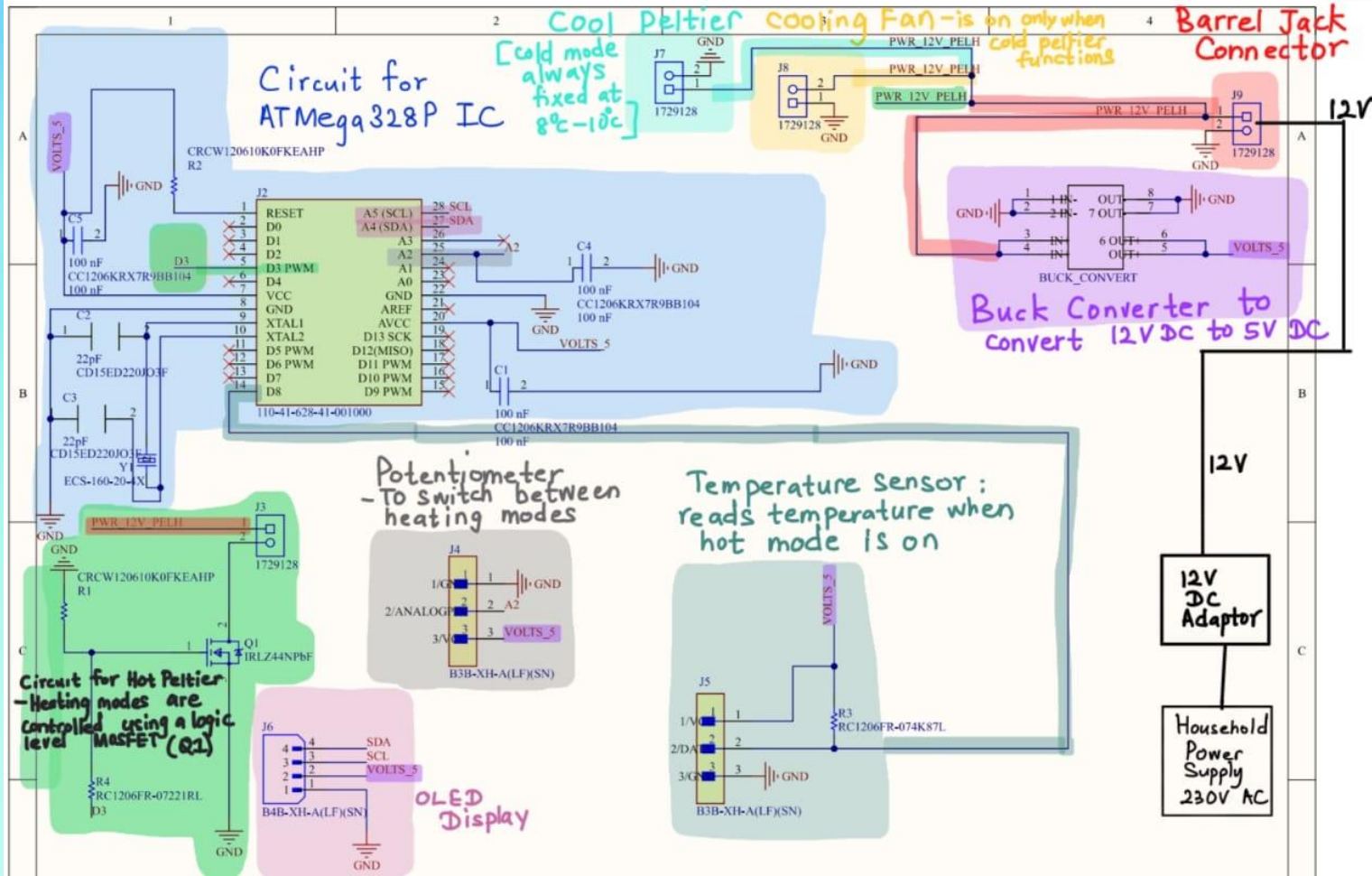
A Peltier module is a solid-state device that heats on one side and cools on the other when electric current passes through it, enabling precise temperature control without moving parts.



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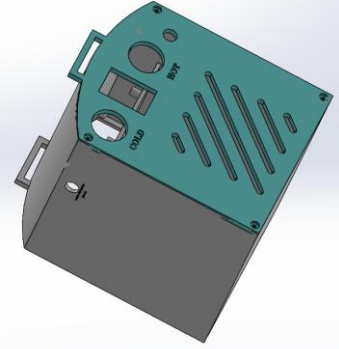
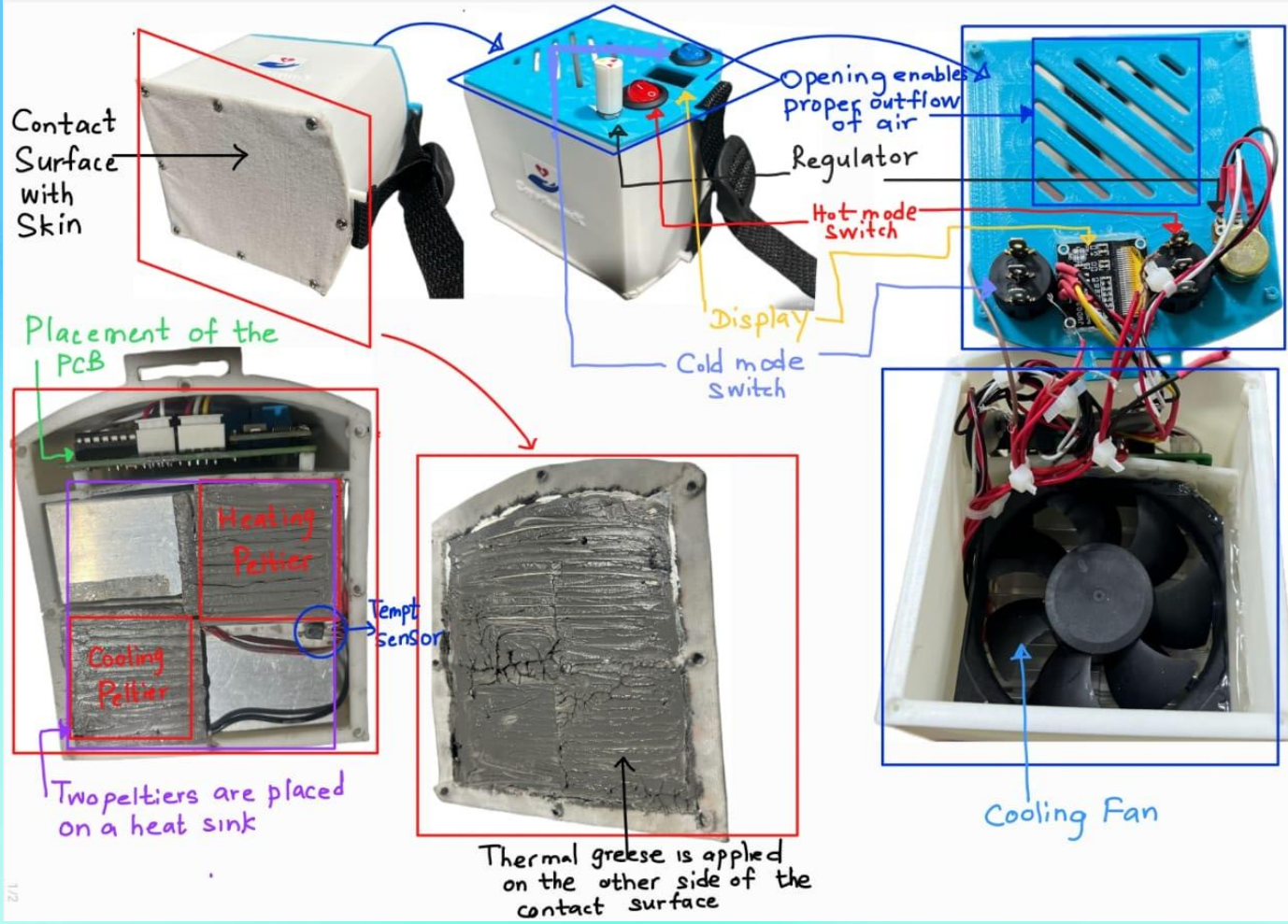
How it works?

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Product Dissection

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Minimum Viable Product (MVP) and Cost

Product CryothermX	
Size	8cm x 9cm x 9cm
Weight	Less than 500g
Default Cold Settings	8 °C - 10 °C
Default Hot Settings	Low - 35 °C
	Medium - 41 °C
	High - 48 °C
Price	Cost: Rs. 12k



Product	Quantity	Price
Peltier 12706	2	1340
IRLZ44N MOSFET	1	120
100nF capacitors	3	30
221 ohm resisors	3	9
4.7K ohm resisors	2	6
Heat sink	2	420
Fan	1	530
JST connectors	4	40
Screw terminal	4	40
Atmega IC	1	1600
Buck converter	2	460
Enclosure print		4150
velcro strip	1	150
metal plate	1	120
Screws	10	30
Thermal paste	1	340
Baral jack connector	1	20
12V/5A adaptor	1	1590
potentiometer	1	20
OLED display	1	650
LED illuminated switches	2	40
DS18B20 temperature sensor	1	120
Total expenses		11825

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Alternatives Analysis

Therm-X AT Hot/Cold Compression Unit



The Therm-X AT combines cold, heat, contrast, and compression therapy, in one easily portable machine.

Designed for use in Rehabilitation and Sports Medicine, the Therm-X is intended to treat post-surgical and acute injuries to reduce edema, swelling, and pain.

The Therm-X is a flexible tool with digital accuracy, when localized thermal therapy (Heat or Cold) is indicated.

• Size: 10" x 9" x 15"

• Weight: 14 lbs.

- Default Cold Settings: 34°F, 45°F, 55°F
- Default Heat Settings: 105°F, 107°F, 110°F
- Cycle Time Length: 3 to 40 Minutes
- Contrast Temperature Settings: 38°F to 105°F
- Garment Edema Pressures: 5, 20, 45, & 70 Hg

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Part No	Description
710706-001	Therm-X AT Hot/Cold Contrast Compression Unit

710706-002	Therm-X Durable Knee Garment
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	Price	Qty	
Review	\$6,095.00	1	ADD TO CART
			ADD TO WISH LIST
Review	\$495.00	1	ADD TO CART

Product CryothermX

Size

8cm x 9cm x 9cm

Weight

Less than 500g

Default Cold Settings

8 °C – 10 °C

Default Hot Settings

Low – 35 °C

Medium – 41 °C

High – 48 °C

Price

Cost: Rs. 12k

Source: <https://www.tru-medical.com/Modalities/Therm-X-AT-Hot-Cold-Compression-Unit>

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Future Scope

1. Use of conductive silver coated fabric as the contact material with the skin

Currently, we use a thin piece of cotton cloth during calibration. However, due to its low thermal conductivity, the cooling effect isn't always felt on the surface—even when the desired low temperature is actually being achieved underneath.

Product link:

<https://www.texcraf-protection.com/emf-fabric/ripstop-silver-conductive-cloth.html>

Ripstop silver conductive cloth



Model No.: **TF-DYL2**

100% Silver Coated Fabric (20% silver / 80% polyamide)

Width: 150 cm

Weight: 80 ± 5 g/sqm

Conductivity: 0.2Ω/cm

Anti-oxidation and anti-perspiration coatings are provided if required.

TF-DYL2 is a 100% silver plated nylon fabric. Conductivity: 0.2Ω/cm, can effectively shield high frequency electromagnetic. It can be used for electronic circuits, 5G base stations, industrial product shielding and household EMF shielding.



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Contribution of Members

230218G	230016K	230581K	230481E
GUNATHUNGA U.A.	ABISHEK L.	SANTHOSH S.	PERERA K.W.A.O.V
1. Microcontroller Programming 2. PCB Designing (mainly)	1. Enclosure implementation using Solidworks (mainly) 2. Soldering	1. Enclosure implementation using Solidworks (mainly) 2. Assembling	1. Initial sketching of design idea(mainly) 2. Circuit Design and Component Selection (mainly) 3. Assembling

Demonstration

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Thank you

Q & A

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